Capstone Proposal Example

BerrySleuth

Business Understanding
I hike frequently and I like eating berries. Often, I eat berries that I find in the woods without knowing exactly what they are. This seems dangerous, but so far I have survived. I would like to create an application that will enable me to predict the species of a local berry plant with just a photo of the plant.

Data Understanding
I will hike into the forest for my capstone project and photograph all the berries. I will need to spend an entire season in the backcountry to make sure that all of the different types of berries are in season at some point during the data collection phase of my project. See you in October!

Data Preparation
My goal is to have anywhere between 300 and 3000 photos per species of berry plant. I'll store them in an S3 bucket with a separate folder for each species. I'll also label them as edible or non-edible. I will write a function to format the images into appropriately pre-processed floating point tensors that can be fed into the network.

Modeling
I'll set aside a final validation set of labeled images, maybe 20% for each class. Then I'll use Tensorflow to build an image classifier by fine tuning a pre-trained base model and `tf.keras.Sequential`. I'll also try using deep feature extraction on the images and making predictions using logistic regression, random forest, etc.

Evaluation
I will report both the accuracy score and cross entropy loss, on training and test data. Because I don't want to die in the woods, I'll tie the evaluation back to my original business problem and choose classification thresholds that maximize recall for poisonous berries. I plan to use k-fold cross validation to do any needed parameter tuning.

Deployment
The model will be deployed as a Flask app that can be used to upload a picture of a plant from a mobile phone to get a prediction of its species.